Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/16/2004
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Project Title:	Individual Susceptibility to Radiation Effects		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Zip Code:	10032	**Congressional District:**	15
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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	The working hypothesis of this proposal was that cells from heterozygous ATM knockout mice are more sensitive to radiation than cells from wild-type mice. Two endpoints related to radiosensitivity were tested. Mouse embryo cells from wild-type and knockout animals were irradiated and the incidence of oncogenic transformation scored; this is an example of a stochastic effect. Radiation induced cataracts were studied in adult wild-type and knockout mice; this represents an important deterministic effect. Oncogenic Transformation Mice heterozygous for the ATM gene were mated, mid-term pregnant animals sacrificed and cell cultures prepared from individual embryos. The cells were genotyped as wild-type, homozygous or heterozygous for the ATM gene. All three cell types (wild-type, homozygous and heterozygous for the ATM gene) were exposed to gamma-rays. There was a		

Task Description:	statistically significant increase in the incidence of oncogenic transformation in cells from AT heterozygotes compared with wild-type cells from litter-mates. Cataractogenesis Wild-type mice, as well as mice heterozygous for the ATM gene were irradiated with graded doses of x-rays and observed regularly for cataract development using both subjective (slit-lamp) biomicroscopy and quantitative objective (Scheimpflug Imaging) measures. Preliminary data indicate that cataracts develop earlier and reach a higher grade in AT heterozygous mice, compared with wild-type animals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The dose limits proposed for astronauts in space flight are different from those on the ground, though the basic philosophy of limiting the life-time risk of stochastic effects to 3% is preserved. There are several reasons for the differences, including the limited number of individuals exposed, the practicality of shielding and the presence of more urgent risks. Nevertheless the purpose of radiation protection remains to prevent deterministic effects of clinical significance and limit stochastic effects to acceptable levels. The levels proposed are based on the assumption that the human population is uniform in its radiosensitivity. Using two endpoints, namely cataracts as an important and relevant deterministic effect and oncogenic transformation in cultured embyro cells as a surrogate for carcinogenesis, the most relevant stochastic effect, we have shown that mice heterozygous for the atm gene are more radiosensitive than wild type animals. 1 to 3% of the US population are AT heterozygotes, and it would not be prudent to subject such individuals to substantial doses of radiation.
Task Progress:	Oncogenic transformation Litter-matched experiments were performed to investigate the influence of atm heterozygosity on radiation-induced oncogenic transformation of MEFs. A total of 13 intralitter comparisons were made between normal and A-T heterozygote embryos. Yields of transformed clones were measured both for zero-dose exposure and for exposure to a gamma-ray dose of 2 Gy. The ATM heterozygous mice were significantly more sensitive for radiation oncogenesis than were the corresponding wild-type animals by a factor of almost 2. Ocular cataracts AT heterozygotes are more radiosensitive than their wild-type litter mates and indeed are intermediate in sensitivity between the wild-type and homozygous animals. At the lowest dose (0.5 Gy) of X-rays the wild-type animal show no evidence of cataracts, while 50% of the AT heterozygotes developed a grade 1 cataract. At the higher doses, (2 and 4 Gy) the AT heterozygotes developed a grade 2.0 cataract about 20 weeks earlier than their wild-type litter mates. Grade 2 corresponds to a clinically significant, vision impairing cataract in the human. Of course the AT homozygote animals developed higher grade cataracts very quickly in their short lives, but this was only to be expected. The question then arose whether heterozygosity might equally influence sensitivity to high LET radiations. The question is not without controversy since the radiosensitive cells referred to above are haploinsufficient for genes involved in DNA repair, and/or checkpoint control. It could be argued that this haploinsufficiency may affect sensitivity to low LET radiation (where much of the damage is repairable) to a greater extent than to high LET radiations (where much of the damage is irreparable anyway). If this were true, the Relative Biological Effectiveness for radiosensitive cells would be smaller than for wild type cells. Together with those of wildtypes, the eyes of AT heterozygous knockout mice were exposed to 0.325 Gy of 1 GeV/amu 56Fe ions at the AGS facility of Brookhaven National Laboratory. The fluence was equivalent to 1 ion per nuclear area. As was the case in the earlier X-ray studies all irradiations were done on the 28th day after birth. Controls consisted of wildtype irradiated as well as unirradiated wildtype and heterozygotes. Ten mice from each group were examined weekly by conventional slitlamp biomicroscopy for a total of 35 weeks. The time for definitive cataract onset (grade 1.0) was shorter in ATM heterozygotes by about 10 weeks than in wild-type animals. At the conclusion of the experiment (35 weeks), 80% of the ATM heterozygotes had progressed to a vision impairing cataract (grade 2.5) compared with 5% of the wild-type. These data assume additional significance in the light of a recent publication linking an increased risk of cataracts for astronauts with higher lens doses (>8 mSv) of space radiation relative to other astronauts with lower lens doses (<8 mSv). In addition, there was a significant association between cataracts and high inclinations or lunar missions, with 35 of the 39 cases observed after space flight occurring in astronauts participating in these missions. It is speculated that the cause is the much higher flux of heavy ions associated with high inclination and lunar missions, with lower inclination missions having a large fraction of the dose from low linear energy transfer trapped protons. Overall, 3 of the 295 astronauts that were followed developed vision-impairing cataracts requiring surgery relatively early in life, even though the accumulated doses were quite low; it is unfortunate that the report does not indicate whether these three cases were of a type consistent with a radiation etiology, but the number possibly suggests a genetically predisposed radiosensitivity in some individuals.
Bibliography Type:	Description: (Last Updated: 10/26/2023)
Articles in Peer-reviewed Journals	Hall EJ. "Henry S. Kaplan Distinguished Scientist Award 2003. The crooked shall be made straight; dose-response relationships for carcinogenesis." Int J Radiat Biol. 2004 May;80(5):327-37. PMID: 15223765 , May-2004
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Articles in Peer-reviewed Journals	Hall EJ, Brenner DJ, Worgul B, Smilenov L. "Genetic susceptibility to radiation." Adv Space Res. 2005;35(2):249-53. PMID: 15934202 , Jan-2005